

Course Code: ITAI1371

Assignment: Lab 05 - Data Preparation

Group Name: Group 01

Lead Submitter & Compiler: Cuong Dang

Date of Submission: June 29, 2026

REFLECTIVE JOURNAL

LAB 05 DATA PREPARATION

I. INTRODUCTION & EXECUTIVE OVERVIEW

Data preparation is widely recognized as one of the most critical and time-consuming phases within the entire Machine Learning workflow. Raw, real-world data is inherently messy, fragmented, and rarely structured in a manner that mathematical algorithms can natively interpret. The primary objective of this laboratory assignment was to practically implement core preprocessing methodologies—specifically data imputation, categorical feature encoding, and feature standardization—using the classic Titanic passenger survival dataset.

As the lead compiler and repository manager for this group assignment, my role expanded beyond individual task execution. I was responsible for asynchronous coordination, code optimization, system debugging, and integrating distinct analytical contributions from five different team members into a single, cohesive end product. This reflective journal encapsulates our collective hurdles, technical discoveries, and institutional growth throughout the lifecycle of Lab 05.

II. TECHNICAL CHALLENGES & SYSTEM DEBUGGING

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The execution of this lab exposed several critical challenges, both in terms of algorithm comprehension and library version compatibility:

1. The Future-Proofing Syntax Crisis (Pandas 3.0 Warning):

While executing the baseline imputation cells in Part 2, our team initially utilized the standard `inplace=True` parameter to substitute missing numerical cells in the 'Age' column. However, the runtime environment generated a prominent pink FutureWarning alert us that chained assignment behaviors are being fundamentally deprecated in upcoming Pandas 3.0 frameworks. Recognizing that this could cause immediate grading compilation failures if the instructor uses an updated interpreter, I took the initiative to refactor the baseline code structure. By transitioning to direct assignment syntax (`df['Age'] = df['Age'].fillna(median_age)`), we successfully eradicated the system warning, ensuring cleaner compiler memory and absolute long-term code stability.

2. Multicollinearity Prevention in Feature Mapping:

During the One-Hot Encoding phase in Part 3, a major conceptual challenge was understanding the strict statistical necessity of the `drop_first=True` argument. Through team collaboration and internal discussion, we verified that generating binary vectors for every single unique category introduces redundant information (the "Dummy Variable Trap"). For instance, keeping both `Sex_female` and `Sex_male` is mathematically

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unnecessary because a 0 in the male column automatically confirms a female passenger.

Dropping the first dummy column keeps our feature space lean and prevents multicollinearity from distorting future model weight vectors.

3. Scale Invariance & Distribution Skewness:

The Titanic dataset contains features on vastly disparate scales; passenger ages range from 0 to 80, while ticket fares range from 0 to 512. If left untreated, distance-based or gradient-descent algorithms would naturally assign disproportionate importance to the Fare metric simply due to its higher numerical range. Implementing StandardScaler allowed us to shift our continuous variables onto a standardized normal distribution with a mean of 0 and a standard deviation of 1, leveling the playing field for proportional feature weightings.

III. GROUP COHESION & WORKFLOW EXPERIMENTS

Managing a five-member team under a compressed timeline and facing asynchronous communication hurdles presented a massive organizational bottleneck. Team collaboration did not flow smoothly, and individual fragments were delivered in isolation. Astrid Lorenzana initiated early trials on Part 1 and Part 2, Jonathan Aldana isolated the encoding cells, Ruben Nuno engineered the standalone scaling parameters, and Dana Bolden drafted rigorous responses for the final theoretical prompts.

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To bridge these gaps and elevate the academic validity of our submission, I formulated an independent **"Part 5: Exploratory Verification Phase."** I wrote three unique data-validation experiments directly into the master pipeline to cross-verify our team's mathematical outputs:

- **Experiment 1** analyzed survival distributions across demographic groups to validate why categorical text encoding is a predictive prerequisite.
- **Experiment 2** checked mean values across ticket tiers to explicitly prove the skewness and outlier presence discussed in our theoretical responses.
- **Experiment 3** printed a statistical summary comparing the raw and scaled matrix boundaries, numerically proving that our scaling operations successfully compressed the dataset's numerical range without destroying underlying variance.

IV. KEY TAKEAWAYS & CRITICAL INSIGHTS

The fundamental takeaway from this lab is that data preparation is not a rigid, linear checklist, but an adaptive, highly analytical art form. Every imputation choice carries immediate downstream consequences. For example, using the mean instead of the median for a skewed metric like Fare would have artificially shifted our center of mass upward due to outlier distortion.

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Furthermore, we have proved that certain architectures, such as Decision Trees, are structurally immune to feature scaling because they partition data points based on independent monotonic splits rather than spatial distance metrics. This realization was a massive breakthrough for our collective understanding of model selection.

Ultimately, despite severe internal team coordination friction, this assignment forced us to develop immense resilience. By acting as the central axis to debug system parameters, run independent validation code, and strictly police the required file-naming and directory structures for GitHub deployment, I learned how to manage messy data—and even messier workflows—to bring a project to successful completion.